

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 - FLOWCHART-TO-CODE CONVERSION

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5 - Naturalization (BL4, BL6)	Assessment:	Assessment Tool 2 - Flowchart-To-Code Conversion
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	K. Kabilan	Reg. No.:	192521157
Section / Batch:	B-Tech-IT	Date:	04/09/2026

Code of Conduct

I K. Kabilan (192521157) (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: K. Kabilan

Instructions

- This test contains 5 Flowchart-to-code conversion questions. Total: 100 Marks.
- When a student answer using Flowchart-to-code conversion should evaluate based on the ability to design clear, logical, and efficient code solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

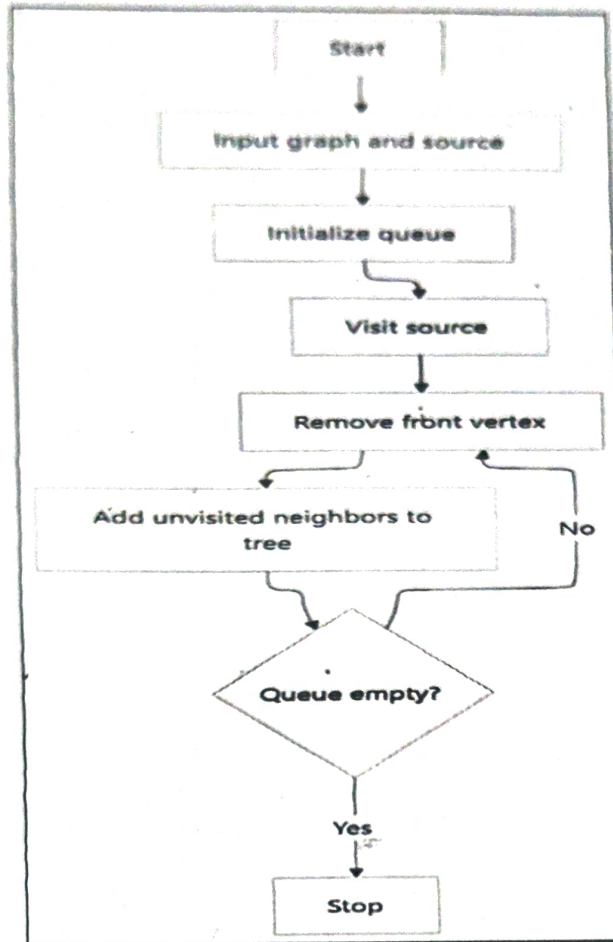
Section / Topic	Focus	Q Nos	Marks
Graph Traversal	BFS	1	20
Shortest Path Algorithm	Dijkstra's Algorithm	2	20
Shortest Path Algorithm	Unweighted Graph	3	20
Graph Traversal	BFS	4	20
Minimum Spanning Tree	Prim's Algorithm	5	20
GRAND TOTAL:			100 Marks

Annexure A - Pseudocode Writing Task

1. Convert the flowchart for generating a BFS tree into code.

(20 Marks)

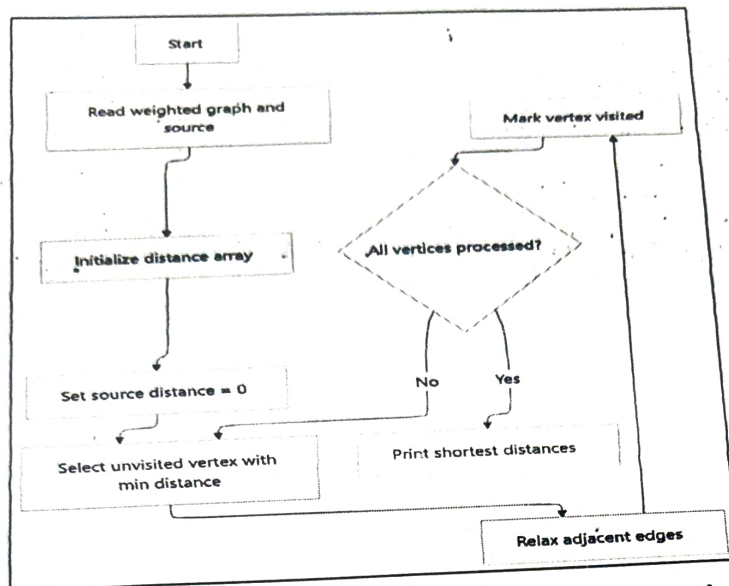
Flowchart Diagram:



2. Convert the flowchart for Dijkstra's shortest path algorithm into code.

(20 Marks)

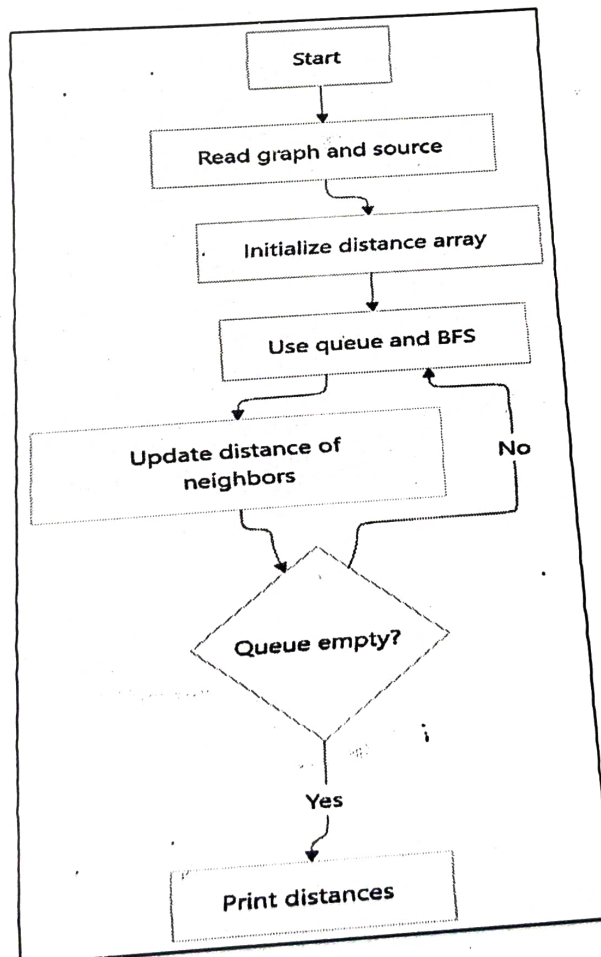
Flowchart Diagram:



3. Convert the flowchart for shortest path in an unweighted graph into code.

(20 Marks)

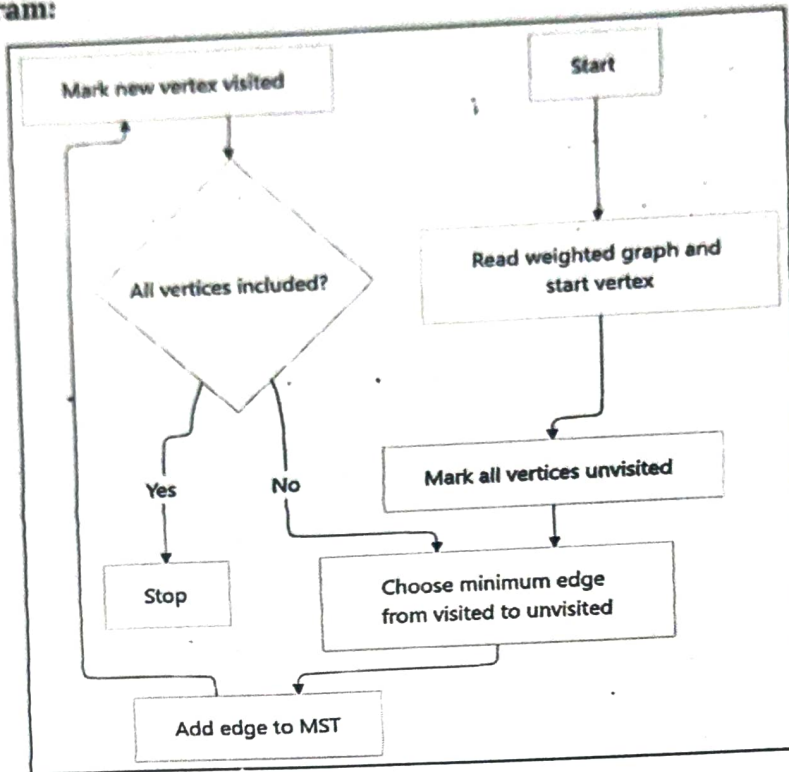
Flowchart Diagram:



4. Convert the flowchart for Prim's algorithm into code.

(20 Marks)

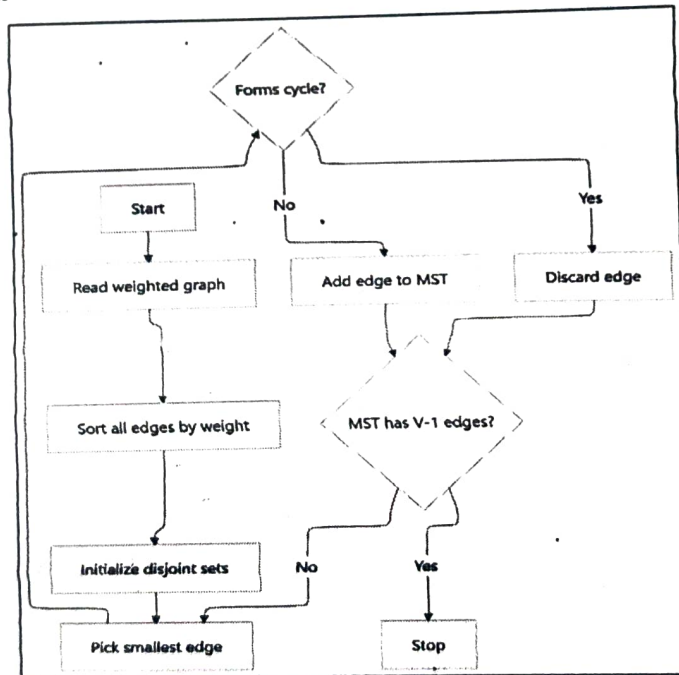
Flowchart Diagram:



5. Convert the flowchart for Kruskal's algorithm into code.

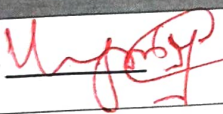
(20 Marks)

Flowchart Diagram:



Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature: 	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Convert the flowchart for generating a BFS tree into code.



Code:

```

#include <stdio.h>
int main() {
    int a[10][10], q[10], v[10] = {0}, k, s, f = 0, r = 0, i, j;
    scanf("%d", &n);
    for(i = 0; i < n; i++) for(j = 0; j < n; j++)
        scanf("%d", &a[i][j]);
    scanf("%d", &r); q[r++] = s; v[s] = 1;
    while(f < r) { s = q[f++]; printf("%d", s);
    for(i = 0; i < n; i++) if(a[s][i] & & v[i] == 0) { q[r++] = i; v[i] = 1; }
    }
}
  
```

2) Convert the flowchart for Dijkstra's into code.



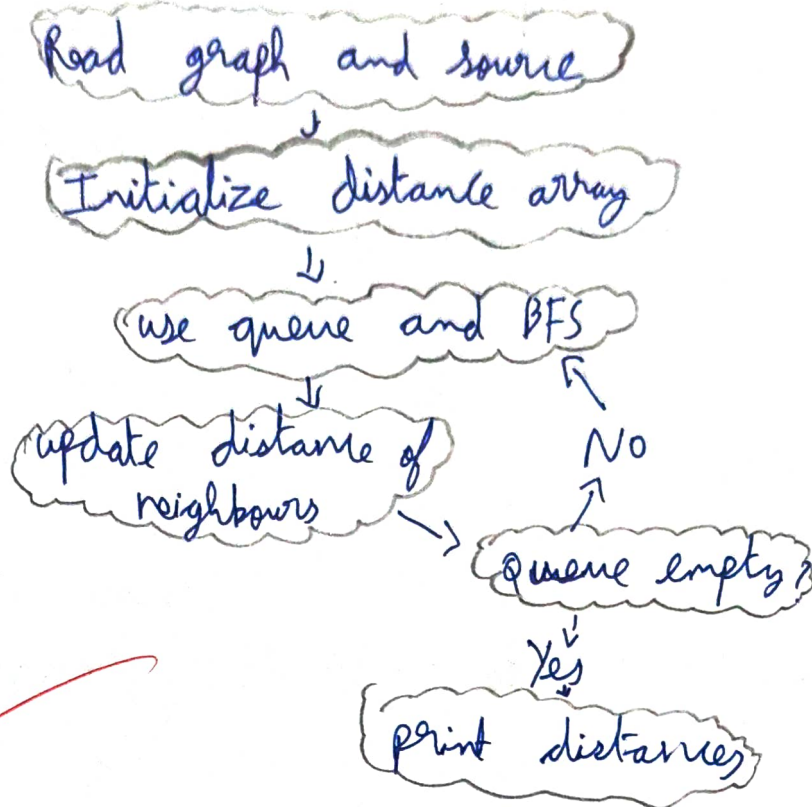
Code:

```

#include <stdio.h>
int main() {
    int a[10][10], d[10], v[10] = {0}, n, s, i, j, v, m;
    scanf("%d", &n);
    for(i=0; i<n; i++) for(j=0; j<n; j++) scanf("%d", &a[i][j]);
    scanf("%d", &s);
    for(i=0; i<n; i++) d[i] = 999;
    d[s] = 0;
    for(i=0; i<n; i++) { m = 999;
        for(j=0; j<n; j++) if(v[j] & & d[j] < m) m = d[j], u = j;
        v[u] = 1;
        for(j=0; j<n; j++) if(a[u][j] & & d[j] > d[u] + a[u][j])
  
```


convert the flowchart for shortest path in an unweighted graph into code.

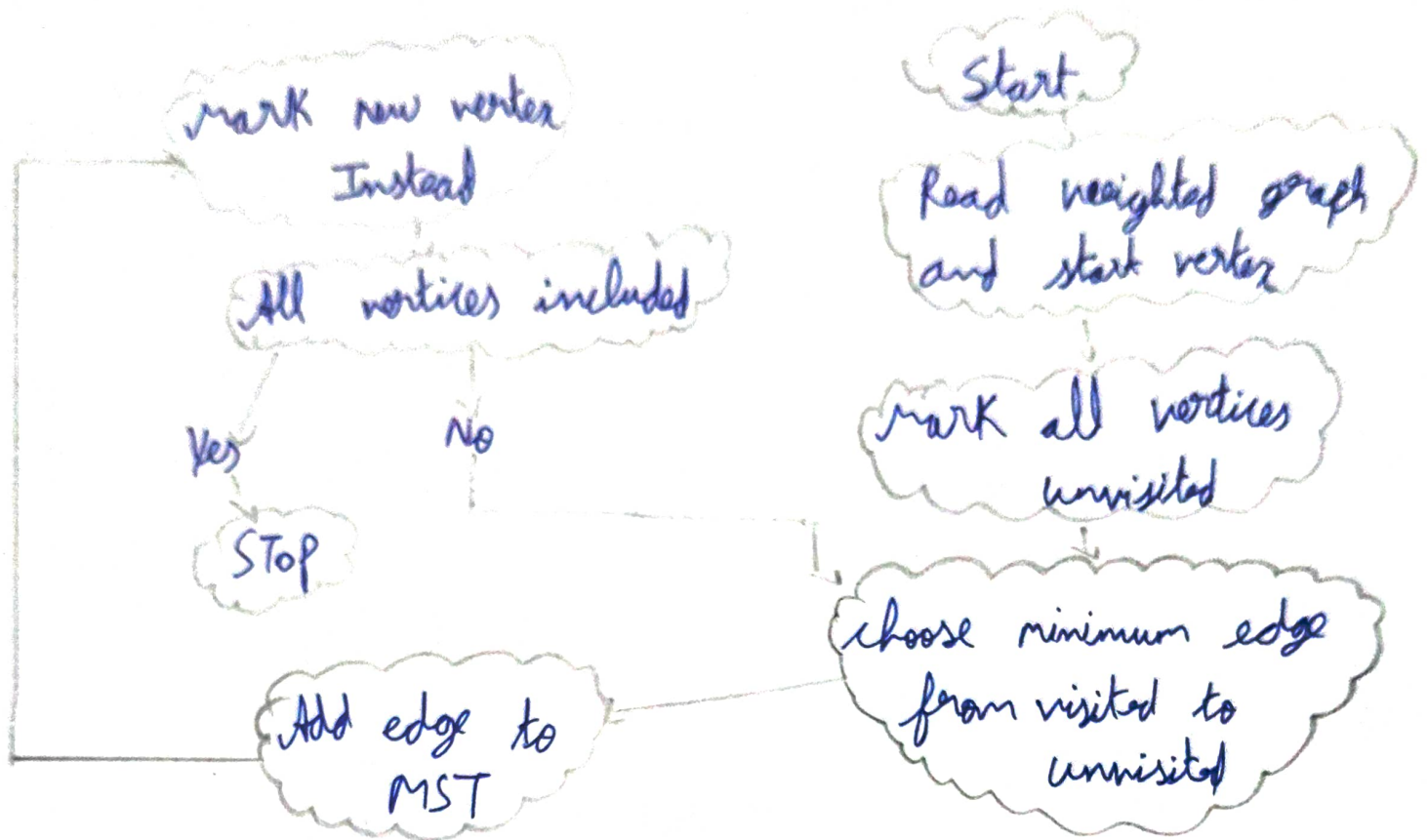
start



code:

```
#include <stdio.h>
int main() {
    int a[10]
    [10], d[10], q[10], v[10] = {0}, 15 n, f = 0, s = 0, u, i, j;
    scanf("%i", &n);
    for (i = 0; i < n; i++) for (j = 0; j < n; j++)
        scanf("%i", &a[i][j]);
    scanf("%i", &s);
    for (i = 0; i < n; i++) d[i] = -1;
    q[s] = s; v[s] = 1; d[s] = 0;
    while (f < n) { u = q[f++];
        for (i = 0; i < n; i++) if (a[u][i] & d[i]
            v[i] & v[i] = 1 d[i] = d[u] + 1, q[n++] = i;
        for (i = 0; i < n; i++) printf("%i", d[i]);
    }
```


t) convert the flowchart for prime algorithm into code.



code:

```

#include <stdio.h>
int main() {
    int a[10][10], V[10] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, x, m, l = 0;
    scanf("%i", &n);
    for (i = 0; i < n; i++) for (j = 0; j < n; j++) scanf("%i", &a[i][j]);
    while (l < n - 1) {
        m = 999;
        for (i = 0; i < n; i++) if (V[i]) for (j = 0; j < n; j++)
            if ((i < j) & &a[i][j] < m) m = a[i][j], x = j;
        V[x] = 1; printf("%i", m); l++;
    }
}
  
```

convert the flowchart for Kruskal's algorithm into



Code:

```

#include <stdio.h>
int P[10];
int f(int x) { return P[x] == x ? x : P[x] = f(P[x]); }
int main() {
    int a[10][10], n, i, j, x, y, m, c = 0;
    scanf("%i %d", &n, &n);
    for(i = 0; i < n; i++) { P[i] = i; for(j = 0; j < n; j++)
        scanf("%i %d", &a[i][j], &a[j][i]); }
    while(c < n - 1) { m = 999;
        for(i = 0; i < n; i++) for(j = i + 1; j < n; j++)
            if(a[i][j] < a[x][y] && a[i][j] < m) m = a[i][j], x = i, y = j;
        x = f(x); y = f(y);
        if(x != y) P[x] = y, printf("%i %d", m), c++;
    }
}
    
```


SIMATC

$$a [x] [y] = a [y] [x] = 1991y$$

y

changed

ode: CS
d: CO4
30% a
Apply h
data.
K-Ka
B-Test

lan (14
on is my or
that any vi
K-Kab

comparati
using co
exity adv
answered
nic dev

M
Qu
Con
Recu